

Completion Check (Week 7) – Law of Demeter & S.O.L.I.D. Principles Exercises

I. Law of Demeter

1. Do the following parts of code violate the Law of Demeter? Argument your answer. If it is violated provide a solution to respect the Law of Demeter.

a.

```
public void printSomePoints(List<Point> points) {  
    int printedPoints = 0;  
    for (Point point : points) {  
        if (someCondition()) {  
            System.out.println(String.format("Point %d (%f,%f)",  
                printedPoints, point.getX(), point.getY()));  
        }  
    }  
}  
  
class Point {  
    private double x;  
    private double y;  
  
    //getters and setters  
}
```

b.

```
public void printDistanceBetweenRandomPointsOfTwoSquares (Square  
s1, Square s2) {  
    Point randomPoint1 = s1.getRandomPoint();  
    Point randomPoint2 = s2.getRandomPoint();  
  
    System.out.println(randomPoint1.distanceTo(randomPoint2));  
}  
  
@AllArgsConstructor  
class Point {  
    @Getter private double x;  
    @Getter private double y;  
  
    public double distanceTo(Point p){  
        return Math.hypot(x-p.x, y-p.y);  
    }  
}
```

```

@Data //getters and setters
@AllArgsConstructor
class Square {
    private Point[] points = new Point[4];

    public Point getRandomPoint(){
        int index = (int) (Math.random()*4);
        return points[index];
    }
}

```

c.

```

public void someMethod(Student student) {
    Faculty faculty = student.getFaculty();

    System.out.println(faculty.getName());
    System.out.println(faculty.getStudentNo());
}

```

```

@Data
@AllArgsConstructor
class Student {
    private String name;
    private Faculty faculty;
}

```

```

@Data
@AllArgsConstructor
class Faculty {
    private List<Student> students;

    public int getStudentNo() { return students.size(); }
    public void removeStudent(Student student) {
students.remove(student); }
    public void addStudent(Student student) {
students.add(student); }
}

```

d.

```

public void someMethod() {
    //...some code
    Student student = Student.builder()
        .lastName("LN")
        .firstName("FN")
        .age(20)
        .bestFriend(null)
}

```

```

        .build();
        System.out.println(student);
        //...some other code
    }

    @AllArgsConstructor
    class Student {
        private String firstName;
        private String lastName;
        private Integer age;
        private Student bestFriend;

        public static StudentBuilder builder() {
            return new StudentBuilder();
        }
    }

    class StudentBuilder {
        private String firstName;
        private String lastName;
        private Integer age;
        private Student bestFriend;

        StudentBuilder() { }

        public StudentBuilder firstName(String firstName) {
            this.firstName = firstName; return this; }

        public StudentBuilder lastName(String lastName) {
            this.lastName = lastName; return this; }

        public StudentBuilder age(Integer age) { this.age = age;
            return this; }

        public StudentBuilder bestFriend(Student bestFriend) {
            this.bestFriend = bestFriend; return this; }

        public Student build() { return new Student(firstName,
            lastName, age, bestFriend); }
    }

```

2. What is the difference between a data structure and a class? To which one does the Law of Demeter apply to and why?